Curriculum Vitae — Jianyu MA

Date of birth: October 26, 1998
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CURRENT POSITION

2025 – 2026	**Temporary Teaching and Research Fellow** at University of Toulouse.
2024 – 2025	**Temporary Teaching and Research Fellow** at ENAC.
2023 – 2024	**PhD Candidate with Teaching Duties** at Paul Sabatier University.
2022 – 2023	**PhD Candidate with Teaching Duties** at Paul Sabatier University.

EDUCATION

2021 – 2025	**PhD Candidate in Mathematics** [Geometry and Optimal Transport] Toulouse Mathematics Institute, University of Toulouse The manuscript is detailed in the *Publications and Works* section.
2020 – 2021	**Master 2**, Research and Innovation, Paul Sabatier University, France
2019 – 2020	**Master 1**, Higher Education and Research, Paul Sabatier University, France
2016 – 2020	**Bachelor's Degree**, Fundamental Mathematics, Wuhan University, China

PUBLICATIONS AND WORKS

My research focuses on the regularity of Wasserstein barycenters and the influence of the geometry of the underlying metric space. I am particularly interested in how properties like a lower bound on the Ricci curvature determine whether a barycenter is absolutely continuous or singular.

Doctoral Thesis

Regularity of Wasserstein Barycenters
Defended on December 16, 2025. Advisor: Jérôme BERTRAND. Theses.fr:2025TLSES113

> **Commentary:** My thesis studies the impact of geometry on the regularity of barycenters by contrasting two frameworks: Riemannian manifolds and metric trees. The first part of this work, which is the subject of publication [1] below, establishes a regularity theorem for barycenters on manifolds. The second part, not yet published, develops a systematic framework for the analysis of barycenters on metric trees. This framework is based on the combination of two methodological tools I developed: an innovative localization principle, the *restriction property*, and a *reduction technique* that reduces transport problems on trees to the simpler setting of the real line. The articulation of these two tools allows for a fine characterization of the structure of barycenters and explains how the branching topology of the tree generates phenomena of singularity and non-uniqueness.

Publication in a peer-reviewed journal

1. **Absolute continuity of Wasserstein barycenters on manifolds with a lower Ricci curvature bound**
   Calculus of Variations and Partial Differential Equations, 65, 9 (2026), arXiv:2310.13832

   > **Commentary:** This article presents the first major contribution of my thesis. It is shown that on a complete Riemannian manifold with a lower Ricci curvature bound, the Wasserstein barycenter is absolutely continuous as long as the defining probability measure assigns a positive mass to the set of absolutely continuous measures. This result simultaneously lifts the hypothesis of the manifold's compactness and the restrictive one of uniform boundedness of densities, present in previous works. The proof is based on new analytical tools, notably a Hessian equality for barycenters and adapted displacement functionals.

TEACHING EXPERIENCE

Temporary Teaching and Research Fellow at University of Toulouse

2025 – 2026	**Tutorial** 24h	Introduction to Real Analysis	(L1 Level, FSI.Math)
	Lecture-Tutorial 18h	Introduction to Statistics	(L2 Level, FSI.BioGéo)
	Tutorial 20h, **Lab** 24h	Random Simulation	(M1 Level, FSI.Math)
	Lab 12h	Probabilities	(M1 Level, FSI.Math)
	Project 9h, **Lab** 18h	Big Data	(M2 Level, FSI.Math)
	Lab 12h	Continuous Probabilities and Statistics	(L3 Level, FSI.Math)
	Lab 12h	Numerical Methods [Stochastic]	(L3 Level, FSI.Math)
	Tutorial 16h, **Lab** 20h	Machine Learning	(L3 Level, FSI.Math)

Temporary Teaching and Research Fellow at ENAC

2024 – 2025	**Lecture** 17h	Measure Theory	(IENAC 1st year)
	Lecture 20h	Stochastic Processes	(IENAC 2nd year)

Teaching Assistant at Paul Sabatier University

2022 – 2023	**Tutorial** 22h, **Lab** 8h	Introduction to Probability Theory	(L2 Level, FSI.Math)
	Tutorial 4h	Advanced Continuous Probabilities and Statistics	(L3 Level, FSI.Math)
2021 – 2022	**Tutorial** 28h, **Lab** 4h	Advanced Continuous Probabilities and Statistics	(L3 Level, FSI.Math)

ORAL COMMUNICATIONS

Talks at conferences, workshops, and seminars

Nov 05, 2024	Young Mathematicians in Geometry and Analysis, Mulhouse
May 21, 2024	PAS Seminar, Blaise Pascal Mathematics Laboratory (Clermont-Ferrand)
Mar 14, 2024	Images, Optimization and Probabilities Seminar, University of Bordeaux
Feb 05, 2024	GT CalVa, Paris-Cité University (Grands Moulins campus)
Jun 26, 2023	Analysis Seminar, Toulouse Mathematics Institute
May 25, 2023	PhD Students Seminar, Toulouse Mathematics Institute

SKILLS

Languages Chinese (Native), English (C1), French (B2)
IT Skills JavaScript, Kotlin/Java, Mathematica, Python, C++
Projects LSPosed, NeoZygisk, TEESimulator, ChromeXt, DIY homepage